R16

Code No: 135BY

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
COMPUTER GRAPHICS
(Common to CE, CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
- - -

1.a) List the operating characteristics for the following display technologies: raster refresh systems, vector refresh systems, plasma panels and CDs.
b) Set up a procedure for a parallel implementation of the midpoint ellipse algorithm. [8+7]

2.a) Write a boundary-fill procedure to fill an 8-connected region.
b) Write a procedure to fill the interior of give ellipse with specified pattern. [8+7]

3.a) Discuss why the Sutherland-Hodgeman polygon clipping algorithm works for only convex clipping regions.
b) Contrast the efficiency of Cohen-Sutherland and Liang-Barsky line clipping algorithms. [8+7]

4.a) Derive the transformation matrix for rotation about an arbitrary point.
b) Explain about the 2D viewing function. [8+7]

5. Briefly explain the Bezier and B-Spline surfaces. [15]

6. Derive the transformation matrix for Hermite curve. [15]

7. Distinguish between object-space and image space methods of visible surface detection algorithms. Give examples for each. [15]

8.a) Explain briefly about key frame systems.
b) Design a storyboard layout and accompanying key frames for an animation of a single polyhedron. [7+8]

---oo0oo---

USED PAPERS SEP-2021